



Accenture Cloud Computing — Tier 2

Tier 2 moves from fundamentals into the services and architecture concepts that commonly appear in technical MCQs and fresher interviews.

1. Cloud Storage ★★★★★

Cloud providers offer different types of storage depending on how the application accesses data. The three important types are:

- **Object Storage**
- **Block Storage**
- **File Storage**

This is a high-value MCQ topic.

2. Object Storage ★★★★★

Object storage stores data as objects. Each object generally contains:

```
Object
├── Data
├── Metadata
└── Unique identifier/key
```

Examples: Images, Videos, Backups, Documents, Logs

```
Photo.jpg
Video.mp4
Backup.zip
    ↓
Object Storage
```

Cloud Examples:

- **AWS example:** Amazon S3
- **Azure example:** Azure Blob Storage
- **Google Cloud example:** Cloud Storage

Best use case: Large amounts of unstructured data.

 **Remember:** Images/videos/backups → Object Storage

3. Block Storage ★★★★★

Block storage divides data into fixed-size blocks. It behaves like a disk attached to a server/VM.

```
VM
↓
Block Storage
↓
Virtual Disk
```

Typical use cases:

- Operating system disks
- Databases
- Applications requiring low-latency disk access

AWS example: Amazon EBS

 **Remember:** VM disk → Block Storage

4. File Storage ★★★★★

File storage organizes data into:

```
Folders
↓
Files
```

It provides a shared filesystem that multiple systems can access.

```
/shared
├─ file1.txt
├─ file2.pdf
└─ images/
```

Useful when applications require traditional file/folder semantics.

AWS example: Amazon EFS

Object vs Block vs File

Type	Basic idea	Common use
Object	Objects + metadata	Images, videos, backups
Block	Virtual disk blocks	VM disks, databases
File	Files/folders	Shared filesystem

Easy memory:

- Object → S3 → Files as objects
- Block → EBS → Disk
- File → EFS → Shared folders

5. Cloud Database ★★★★★

Cloud providers offer managed database services.

Two broad categories you should know: **Relational** and **NoSQL**.

Relational Database

Data is stored in Tables.

```
Students
ID | Name | Course
1  | Amit | CSE
2  | Ravi | ECE
```

Common technologies: MySQL, PostgreSQL, Oracle, SQL Server.

Cloud examples: Amazon RDS, Azure SQL Database, Google Cloud SQL.

NoSQL Database

NoSQL databases use models such as: Document, Key-value, Wide-column, Graph.

```
Example document:
{
  "name": "Abhishek",
  "course": "CSE"
}
```

Cloud examples: Amazon DynamoDB, Azure Cosmos DB, Google Firestore.

When to use?

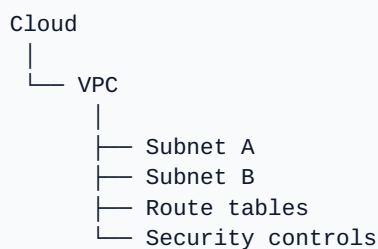
Relational: When you need structured data, SQL, relationships, and strong transactional requirements.

NoSQL: Useful for flexible schemas, very large-scale workloads, and certain high-throughput applications.

6. VPC / Virtual Network ★★★★★

This is an important cloud networking concept.

VPC = Virtual Private Cloud: It is a logically isolated virtual network in a public cloud.

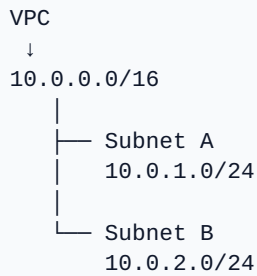


- AWS calls it: **VPC**
- Azure commonly uses: **Virtual Network (VNet)**

Why VPC? You can define IP ranges, Subnets, Routing, Network security, and Connectivity. It's like creating your own logically isolated network inside the cloud.

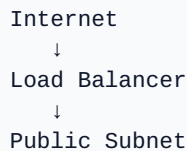
7. Subnet ★★★★★

A subnet is a subdivision of a VPC/VNet IP address range.

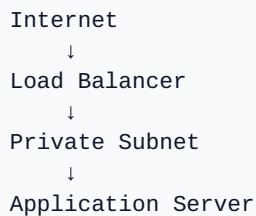


Subnets are commonly categorized as: **Public subnet** and **Private subnet**.

Public Subnet: A subnet whose routing configuration allows resources to have a path to/from the Internet, typically through an Internet gateway or equivalent architecture.



Private Subnet: A subnet without direct inbound Internet connectivity through an Internet gateway.

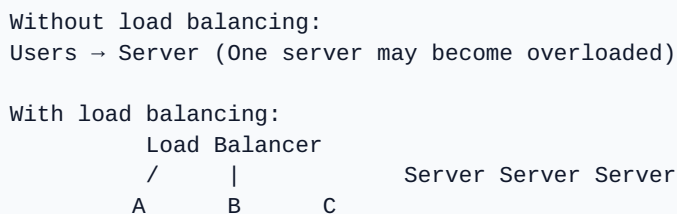


A private subnet can still have outbound Internet access through a NAT gateway/NAT device, depending on the architecture.

 **MCQ trap:** Private subnet does NOT necessarily mean "no Internet access at all." It generally means: *No direct inbound Internet route to the resources.*

8. Load Balancer ★★★★★

Definition: A load balancer distributes incoming traffic across multiple backend servers.



Benefits:

1. **Load distribution:** Traffic is distributed among servers.
2. **High availability:** If one server fails, traffic can be sent to healthy servers.
3. **Scalability:** Works well with multiple application instances.
4. **Health checks:** The load balancer can detect unhealthy backend servers.

9. Auto Scaling ★★★★★

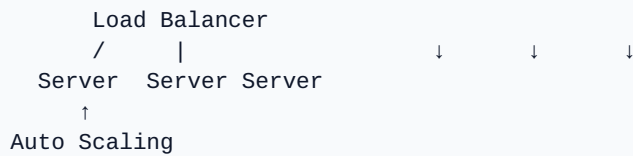
Auto scaling automatically adjusts the number of computing resources based on demand or defined policies.

Normal: 2 servers → Traffic ↑: 5 servers → Traffic ↓: 2 servers

Important distinction:

- **Load balancer:** Distributes traffic.
- **Auto scaling:** Adjusts number of resources.

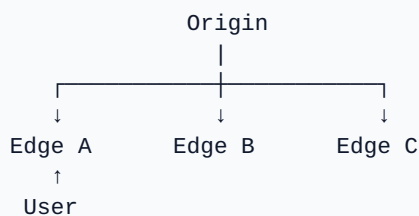
They are often used together:



10. CDN ★★★★★

CDN = Content Delivery Network

A CDN stores/caches content at geographically distributed edge locations so users can retrieve it from a location closer to them.



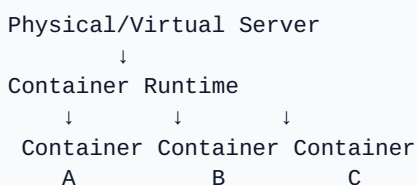
Why CDN? Suppose your server is in India and a user is in the USA. Instead of always requesting static content from India (USA User → India Server), the content may be served from a nearby edge location (USA User → Nearby Edge). This can reduce latency and improve performance.

Best for: Images, CSS, JavaScript, Videos, Static files.

 **Remember:** CDN → Content closer to users

11. Container ★★★★★

A container packages an application with its required dependencies in an isolated environment while sharing the host OS kernel.



Why containers? Lightweight, Portable, Fast to start, Useful for microservices, Consistent across environments.

12. Docker ★★★★★

Docker is a platform/tooling ecosystem widely used to build, package, and run containerized applications.

Application + Dependencies → Docker Image → Container

Docker Image: An image is a packaged template used to create containers.

Docker Image → Container 1, Container 2, Container 3

Important: Image = Template | Container = Running instance

13. VM vs Container ★★★★★

This is a very common interview/MCQ comparison.

VM Architecture:

Hardware → Hypervisor → VM → Guest OS → Application (Each VM has its own guest OS)

Container Architecture:

Hardware → Host OS → Container Runtime → Containers → Applications (Containers share host OS kernel)

Attribute	Virtual Machine (VM)	Container
Operating System	Includes guest OS	Shares host OS kernel
Weight	Heavier	Lightweight
Startup Time	Usually slower startup	Usually faster startup
Isolation	Stronger isolation boundary in many designs	Process-level / application isolation
Overhead	More resource overhead	Lower overhead

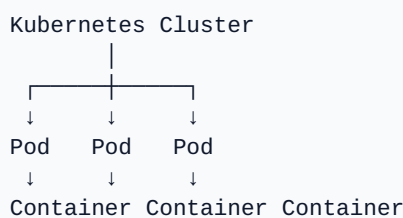
 **Remember:** VM → OS included | Container → Shares host kernel

14. Kubernetes ★★★★★

Definition: Kubernetes is an open-source platform for orchestrating containerized applications.

In simple words: Docker helps run containers; Kubernetes helps manage many containers.

What Kubernetes can do: Deploy containers, Scale applications, Restart failed containers, Service discovery, Load balancing, Rolling updates.



Pod: The basic deployable unit in Kubernetes is Pod. A pod can contain one or more containers that share certain resources/networking.

For beginner MCQs: Pod → Smallest deployable unit in Kubernetes

15. IAM ★★★★★

IAM = Identity and Access Management

IAM controls: Who can access what resources and what actions they can perform.

User → IAM → Permissions → Cloud Resource

IAM Components: Users, Groups, Roles, Policies, Permissions.

Developer → Role → Read S3 → Allowed

Principle of Least Privilege ★★★★★: Give users/services only the permissions they actually need.

Example: A developer only needs to read files from a storage bucket. Don't give full administrator access; give read-only access to that resource.

🧠 **Least privilege** = Minimum required permissions

16. Authentication vs Authorization ★★★★★

Very common interview question.

- **Authentication:** Answers "Who are you?" (e.g., Username + Password → Authentication → Identity verified)
- **Authorization:** Answers "What are you allowed to do?" (e.g., Authenticated User → Authorization → Can read DB? YES; Can delete DB? NO)

🔥 **Easy memory:**

- Authentication = Who are you?
- Authorization = What can you do?

17. Encryption ★★★★★

Cloud data can be protected using encryption. Two important states:

- **Data at Rest:** Data stored somewhere (Disk, Database, Object storage). Database → Encrypted data
- **Data in Transit:** Data moving across a network. Client → Encrypted connection (TLS) → Server

18. Shared Responsibility Model ★★★★★

This is a very important cloud security concept. Cloud security is a shared responsibility between: **Cloud provider + customer**. The exact division depends on the service.

- **Provider generally responsible for:** Physical data centers, Physical servers, Physical networking, Underlying cloud infrastructure.
- **Customer generally responsible for:** Data, Identities/permissions, Application security, Configuration, Guest OS in IaaS.

Example (IaaS VM):

- Cloud provider: Physical hardware, Hypervisor, Cloud infrastructure.
- Customer: Guest OS, Applications, Data, IAM/configuration.

🧠 **Important:** Cloud provider secures *the cloud* infrastructure; Customer is responsible for security *in the cloud*.

19. Managed Services ★★★★★

A managed cloud service means the cloud provider handles much of the underlying operational management.

Your Application → Managed Database → Cloud Provider manages infrastructure

Examples: Amazon RDS, Azure SQL Database, Google Cloud SQL.

Benefit: You spend less time managing infrastructure.

20. Region and Availability Zone ★★★★★

These concepts are important when discussing cloud availability.

- **Region:** A region is a geographic area containing cloud infrastructure (e.g., India Region, USA Region, Europe Region).
- **Availability Zone:** An Availability Zone is an isolated location/infrastructure grouping within a cloud region, designed to provide failure isolation.

Region

- AZ 1
- AZ 2
- AZ 3

Deploying across multiple AZs can improve availability.

 **Remember:** Region → Geographic area | Availability Zone → Isolated location within a region

Accenture Cloud Computing — Tier 2 MCQs

Practice multiple choice questions with detailed explanations and memory tips placed immediately after each question.

Q1. Object Storage ★★★★★

A company wants to store millions of images, videos, and backup files. The data does not need to behave like a traditional disk. Which storage type is most suitable?

- A. Block storage
- B. Object storage
- C. CPU cache
- D. Register storage

 **Answer: B. Object storage**

- A  **Block storage:** Better suited to virtual disks, operating systems, and databases.
- B  **Object storage:** Designed for objects such as images, videos, documents, and backups.
- C  **CPU cache:** Very small, high-speed memory used by processors.
- D  **Register storage:** Tiny storage inside the CPU.

 Images/videos/backups → Object storage

Q2. Block Storage

A cloud VM needs a persistent virtual disk on which to install an operating system. Which storage is most appropriate?

- A. Object storage
- B. Block storage
- C. CDN
- D. DNS

✓ **Answer: B. Block storage**

- A ✗: Object storage is not normally used as the VM's traditional boot disk.
- B ✓: Block storage behaves like a virtual disk attached to a VM.
- C ✗: CDN distributes/caches content.
- D ✗: DNS resolves names.

🧠 VM disk → Block storage

Q3. File Storage

Several application servers need to access the same directory containing files. Which storage model is most suitable?

- A. File storage
- B. Block storage only
- C. CPU cache
- D. Object storage only

✓ **Answer: A. File storage**

- A ✓: Provides a shared filesystem with files/directories.
- B ✗: Block storage generally presents individual block devices rather than a shared filesystem by itself.
- C ✗: CPU cache is not persistent shared storage.
- D ✗: Object storage uses objects rather than traditional shared filesystem semantics.

🧠 Shared folders/files → File storage

Q4. Storage Matching

Which option is correctly matched?

- A. Object storage → VM boot disk
- B. Block storage → Images and videos at massive scale
- C. File storage → Shared filesystem
- D. CDN → Database disk

✓ **Answer: C. File storage → Shared filesystem**

- A ✗: VM boot disks commonly use block storage.
- B ✗: Images/videos are common object-storage workloads.
- C ✓: File storage provides filesystem semantics.
- D ✗: CDN caches/distributes content; it isn't a database disk.

Q5. Relational Database ★★★★★

Which database model primarily stores structured data in tables consisting of rows and columns?

- A. Relational database
- B. Object storage
- C. DNS
- D. CDN

✓ **Answer: A. Relational database**

- A ✓: Relational databases use tables, rows, columns, keys, and SQL.
- B ✗: Object storage stores objects.
- C ✗: DNS maps names to network addresses.
- D ✗: CDN distributes/caches content.

Examples: MySQL, PostgreSQL, SQL Server, Oracle

Q6. NoSQL

An application needs a database with a flexible schema and very large-scale distributed workloads. Which option may be more suitable?

- A. NoSQL database
- B. Relational database only
- C. DNS
- D. File storage

✓ **Answer: A. NoSQL database**

- A ✓: NoSQL databases can provide flexible data models and horizontal scalability for appropriate workloads.
- B ✗: Relational databases can scale significantly too, but the scenario points toward a common NoSQL use case.
- C ✗: DNS is a naming system.
- D ✗: File storage is not a database.

🧠 Flexible schema / certain massive-scale workloads → NoSQL

Q7. VPC ★★★★★

What is a VPC?

- A. Virtual Private Cloud
- B. Virtual Processing Cache
- C. Variable Public Computer
- D. Virtual Protocol Controller

✓ **Answer: A. Virtual Private Cloud**

- A ✓: A logically isolated virtual network in a cloud environment.
- B ✗
- C ✗
- D ✗

🧠 VPC = Your logical network in the cloud

Q8. VPC Purpose

Which of the following can be configured within a VPC?

- A. Subnets and routing
- B. CPU instruction set only
- C. Keyboard layout
- D. Monitor resolution

✓ **Answer: A. Subnets and routing**

- A ✓: VPCs provide networking constructs such as IP ranges, subnets, and routing.
- B ✗: CPU architecture isn't the primary purpose of a VPC.
- C ✗: Not a cloud networking function.
- D ✗: Not a VPC function.

Q9. Subnet

A VPC uses the address range 10.0.0.0/16. An administrator creates 10.0.1.0/24 and 10.0.2.0/24 inside it. What are these smaller networks?

- A. Regions
- B. Subnets
- C. Availability Zones
- D. Containers

✓ **Answer: B. Subnets**

- A ✗: Region is a geographic cloud area.
- B ✓: Subnets divide a VPC's IP address space.
- C ✗: AZ is an infrastructure location within a region.
- D ✗: Containers are application runtime environments.

Q10. Private Subnet ★★★★★

A company wants application servers that should not accept direct inbound Internet connections. Where should they generally be placed?

- A. Private subnet
- B. Public subnet only
- C. DNS zone
- D. CDN edge

✓ **Answer: A. Private subnet**

- A ✓: Private subnets are commonly used for internal application/database resources without a direct inbound Internet route.
- B ✗: Public subnets are designed for resources requiring public Internet routing.
- C ✗: DNS zones manage name resolution.
- D ✗: CDN edges cache content.

⚠ **Important:** Private subnet doesn't necessarily mean zero Internet access. Outbound access can be provided through NAT, depending on architecture.

Q11. Public Subnet

A load balancer must accept requests from users on the Internet. Which type of subnet is commonly used for its Internet-facing interfaces?

- A. Public subnet
- B. Private subnet only
- C. CPU subnet
- D. Database subnet

✓ **Answer: A. Public subnet**

- A ✓: An Internet-facing load balancer commonly has public connectivity.
- B ✗: A private subnet has no direct inbound Internet route.
- C ✗: Not a networking concept.
- D ✗: "Database subnet" isn't a standard subnet type.

Q12. Load Balancer ★★★★★

A website receives millions of requests. The company has five application servers and wants incoming requests distributed among them. What should it use?

- A. Load balancer
- B. DNS only
- C. Hypervisor
- D. Object storage

✓ **Answer: A. Load balancer**

- A ✓: Distributes traffic among backend servers.
- B ✗: DNS resolves names and can participate in traffic distribution architectures, but it isn't the primary load-balancing mechanism described here.
- C ✗: Hypervisor manages VMs.
- D ✗: Object storage stores data.

🧠 Distribute incoming traffic → Load balancer

Q13. Load Balancer Health Check

One application server becomes unhealthy. The load balancer detects this and stops sending new requests to it. Which feature is being demonstrated?

- A. Health checking
- B. Object storage
- C. DNS resolution
- D. Encryption at rest

✓ **Answer: A. Health checking**

- A ✓: Load balancers can monitor backend health.
- B ✗: Storage concept.
- C ✗: DNS concept.
- D ✗: Protects stored data.

Q14. Auto Scaling ★★★★★

A web application automatically increases its server instances from 2 to 10 when CPU utilization becomes high. What is this?

- A. Auto scaling
- B. CDN caching
- C. DNS resolution
- D. Static routing

✓ **Answer: A. Auto scaling**

- A ✓: Automatically changes resource capacity based on demand/policies.
- B ✗: CDN caches content.
- C ✗: DNS resolves names.
- D ✗: Routing determines network paths.

Q15. Load Balancer vs Auto Scaling

Which statement is correct?

- A. Load balancer adds servers; auto scaling distributes traffic
- B. Load balancer distributes traffic; auto scaling adjusts resource capacity
- C. Both only store files
- D. Both are database technologies

✓ **Answer: B**

- A ✗: Roles are reversed.
- B ✓: Correct distinction.
- C ✗
- D ✗

- Load Balancer → Where traffic goes
- Auto Scaling → How many resources exist

Q16. CDN ★★★★★

A website wants users around the world to receive images and videos from locations closer to them. Which technology is most appropriate?

- A. CDN
- B. IAM
- C. Hypervisor
- D. DHCP

✓ **Answer: A. CDN**

- A ✓: CDN caches/distributes content through geographically distributed edge locations.
- B ✗: IAM controls identity and permissions.
- C ✗: Hypervisor manages VMs.
- D ✗: DHCP assigns network configuration.

- 🧠 Content closer to users → CDN

Q17. Docker Image

What is a Docker image?

- A. A running container
- B. A packaged template used to create containers
- C. A physical server
- D. A cloud region

✓ Answer: B

- A ✗: A running instance is a container.
- B ✓: Image is a packaged, immutable template used to create containers.
- C ✗: Physical server is infrastructure.
- D ✗: Region is geographic infrastructure.

🧠 Image = Template

Q18. Container

Which statement best describes a container?

- A. A complete physical server
- B. An isolated application environment that shares the host OS kernel
- C. A cloud region
- D. A DNS record

✓ Answer: B

- A ✗: Container isn't physical hardware.
- B ✓: Containers isolate applications while sharing the host kernel in the standard container model.
- C ✗: Region is infrastructure geography.
- D ✗: DNS record is naming information.

Q19. VM vs Container ★★★★★

Which statement is generally correct?

- A. Containers always contain a complete guest OS
- B. VMs generally include a guest OS, while containers share the host OS kernel
- C. VMs cannot run applications
- D. Containers require a separate physical server for each application

✓ Answer: B

- A ✗: Containers normally share the host kernel.
- B ✓: Correct fundamental distinction.
- C ✗: VMs run applications.
- D ✗: Multiple containers can run on the same host.

🧠 VM → Guest OS

🧠 Container → Shared kernel

Q20. Kubernetes ★★★★★

What is Kubernetes primarily used for?

- A. Container orchestration
- B. DNS name registration only
- C. Physical cable management
- D. Database normalization

✓ **Answer: A. Container orchestration**

- A ✓: Kubernetes manages deployment, scaling, networking, and lifecycle of containerized workloads.
- B ✗: DNS is only one supporting capability in some deployments.
- C ✗
- D ✗

🧠 Docker → Containers

🧠 Kubernetes → Manage many containers

Q21. Kubernetes Pod

What is the basic deployable unit in Kubernetes?

- A. Region
- B. Pod
- C. VPC
- D. Bucket

✓ **Answer: B. Pod**

- A ✗: Region is cloud geography.
- B ✓: Pod is Kubernetes' basic deployable unit. (A pod can contain one or more containers).
- C ✗: VPC is a virtual network.
- D ✗: Bucket is commonly associated with object storage.

Q22. IAM ★★★★★

What is the primary purpose of IAM?

- A. Manage identities and access permissions
- B. Store images
- C. Increase CPU speed
- D. Cache videos

✓ **Answer: A. Manage identities and access permissions**

- A ✓: IAM controls who can access resources and what actions they can perform.
- B ✗: Object storage.
- C ✗: Hardware/resource scaling.
- D ✗: CDN.

🧠 IAM → Who can access what?

Q23. Authentication vs Authorization ★★★★★

A user enters a username and password. The system verifies that the credentials belong to that user. Which process is this?

- A. Authorization
- B. Authentication
- C. Encryption at rest
- D. Load balancing

✓ **Answer: B. Authentication**

A ✗ **Authorization:** Determines what an authenticated user is allowed to do.

B ✓ **Authentication:** Verifies identity.

C ✗: Protects stored data.

D ✗: Distributes traffic.

🧠 Authentication → Who are you?

Q24. Authorization

An authenticated employee can read a database but cannot delete records. Which concept determines this?

- A. Authentication
- B. Authorization
- C. Virtualization
- D. CDN

✓ **Answer: B. Authorization**

A ✗: Authentication verifies identity.

B ✓: Authorization determines permitted actions.

C ✗: Creates virtual resources.

D ✗: Content delivery.

🧠 Authorization → What are you allowed to do?

Q25. Least Privilege ★★★★★

A developer only needs permission to read files from a cloud storage bucket. The administrator grants only read permission. Which security principle is being applied?

- A. Least privilege
- B. Maximum privilege
- C. Open access
- D. Resource pooling

✓ **Answer: A. Least privilege**

A ✓: Gives only the permissions required.

B ✗: Opposite of least privilege.

C ✗: Excessive access is insecure.

D ✗: Resource pooling is a cloud characteristic.

🧠 Minimum required permissions → Least privilege

Q26. Encryption at Rest

A cloud database is encrypted while stored on disk. What type of protection is this?

- A. Encryption in transit
- B. Encryption at rest
- C. Authentication
- D. Load balancing

✓ **Answer: B. Encryption at rest**

- A ✗: Protects data while moving across a network.
- B ✓: Protects stored data.
- C ✗: Verifies identity.
- D ✗: Distributes traffic.

Q27. Encryption in Transit

A user connects to a website over HTTPS and the communication is protected using TLS. What is being protected?

- A. Data at rest
- B. Data in transit
- C. CPU registers
- D. Database schema

✓ **Answer: B. Data in transit**

- A ✗: Stored data.
- B ✓: Data moving between systems.
- C ✗
- D ✗

- 🧠 Moving → Transit
- 🧠 Stored → Rest

Q28. Shared Responsibility Model ★★★★★

In a cloud environment, who is responsible for security?

- A. Only the cloud provider
- B. Only the customer
- C. Both provider and customer, with responsibilities depending on the service
- D. Neither

✓ **Answer: C**

- A ✗: Customer still has responsibilities.
- B ✗: Provider secures underlying cloud infrastructure.
- C ✓: Security responsibility is shared.
- D ✗: Both have responsibilities.

Example (IaaS VM):

- Provider: Physical infrastructure + hypervisor
- Customer: Guest OS + application + data + access configuration

Q29. Region vs Availability Zone ★★★★★

Which statement is correct?

- A. A region is an individual VM
- B. An Availability Zone is a geographic country
- C. A region is a geographic cloud area that can contain multiple Availability Zones
- D. Region and Availability Zone always mean exactly the same thing

✓ **Answer: C**

- A ✗: VM is a compute resource.
- B ✗: An AZ is an isolated infrastructure location within a region.
- C ✓: Correct.
- D ✗: They are different concepts.

Region

- ├─ AZ 1
- ├─ AZ 2
- └─ AZ 3

Q30. 🔥 Mixed Scenario

An e-commerce company has the following architecture:

- Users access an Internet-facing load balancer.
- Application servers are in private subnets.
- Images are stored in object storage.
- Multiple application servers are automatically created during traffic spikes.
- IAM controls which employees can access resources.

Which statement is correct?

- A. Load balancer distributes traffic, private subnets protect application servers from direct inbound Internet access, object storage stores images, and auto scaling adjusts capacity
- B. Load balancer stores images, object storage distributes traffic, and IAM performs auto scaling
- C. Private subnets must have no Internet connectivity of any kind
- D. IAM is responsible only for increasing CPU capacity

✓ **Answer: A**

- A ✓ **Correct:** Every component matches its role.
- B ✗: Functions are incorrectly assigned.
- C ✗: Private resources can have outbound Internet connectivity through mechanisms such as NAT; the key point is no direct inbound Internet route.
- D ✗: IAM manages identities and permissions.